

Generation Interconnection System Impact Study Report

For Queue Project Q19-1135

Bullmoose 138 kV

February 23, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1135. The Interconnection Customer proposes a 146.5 MW solar photovoltaic facility interconnecting at the Bullmoose 138 kV substation in the Pennsylvania area. The request was received on January 12, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 428 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$3,095,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Bullmoose - Sumner 138.0 kV L	Bullmoose 345/1 kV Auto	121.8	258	49.7
Quarry - Geigy Substation 138.0 kV	Bullmoose 345-1 kV Auto	114.6	195	32.3
Quarry - Riparia 138.0 kV Line	Bullmoose 345/1kV XFMR	101.8	239	32.3
Riparia - Rec Laprairie 138.0 kV Line	Bullmoose 345/1kV TX	100.4	265	32.3

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Bullmoose - Sumner 138.0 kV	Rebuild line with higher-capacity conductor	\$2,018,000	88% (\$1,784,000)
Quarry - Geigy 138.0 kV	Rebuild line with higher-capacity conductor	\$1,308,000	62% (\$809,000)
Quarry - Riparia 138.0 kV Line	Reconductor line	\$347,000	62% (\$215,000)
Riparia - Rec Laprairie 138.0 kV	Reconductor line	\$463,000	62% (\$287,000)
	Total allocated to project		\$3,095,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.